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Multi-Cloud data platform integration with OneLake Shortcut and Caching

Riccardo Perico
Analytics & AI Architect @ Lucient | Data Platform MVP

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Analytics & AI Architect @ Lucient
+15 years in Microsoft “Data Realm”
Giving my little contribution to the community

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“OneLake is the least “shiny” Microsoft Fabric component but it’s most the precious”

Riccardo Perico ☺



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What is it OneLake?

OneLake for all data

“The OneDrive for data”



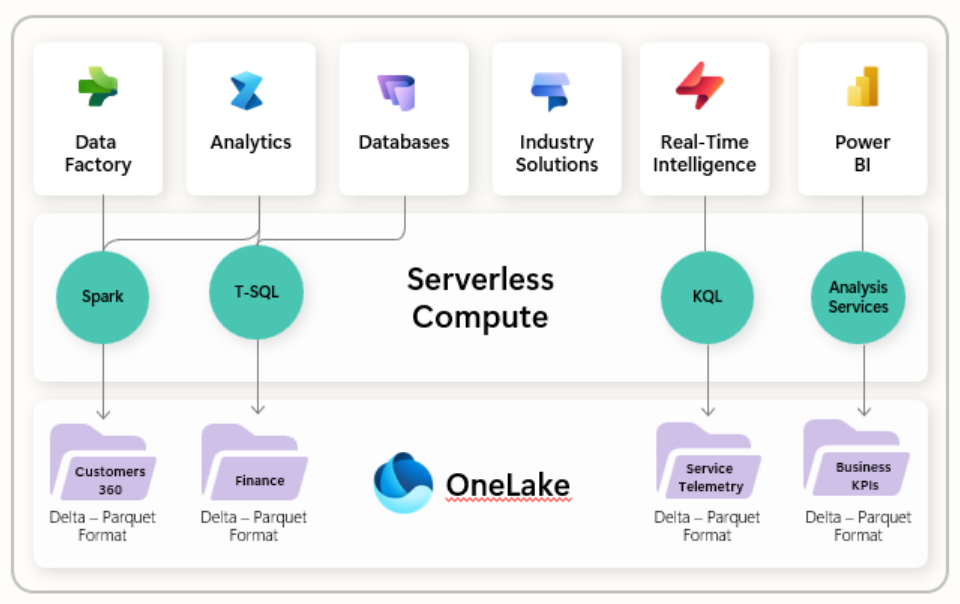
OneDrive



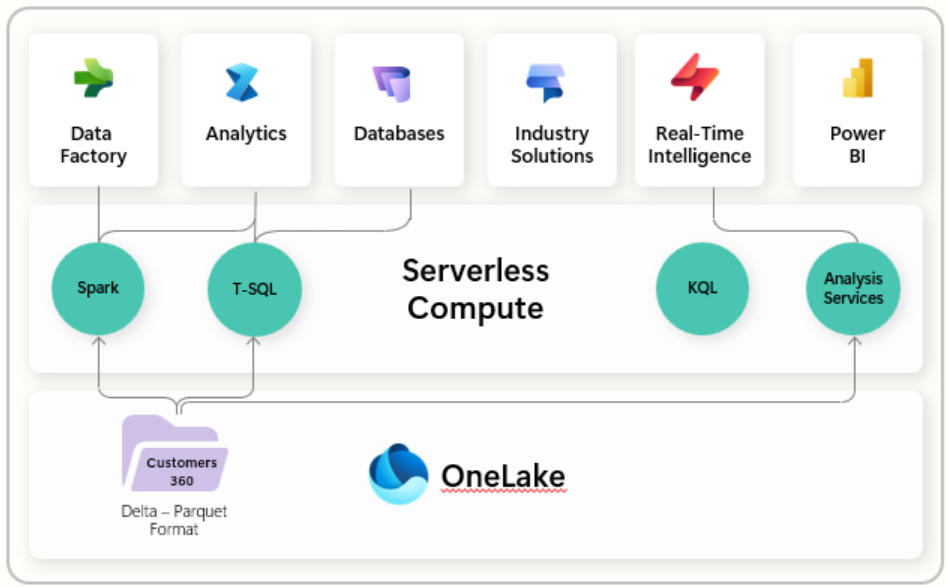
OneLake

OneLake provides a data lake as a service without you needing to build it

All Fabric Workloads can write in it



All Fabric Workloads can read from it



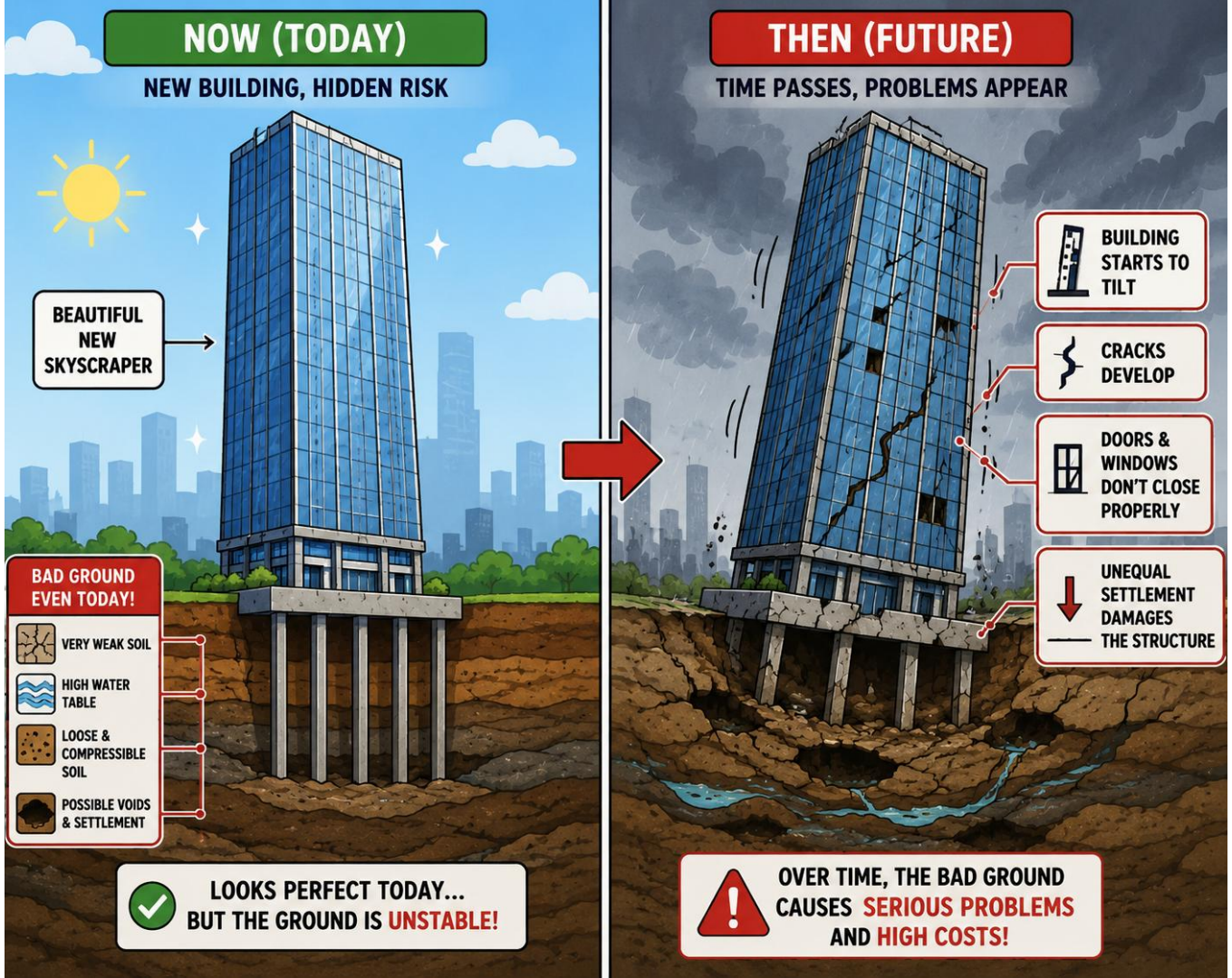
Revolutionary decision by Microsoft



ICEBERG

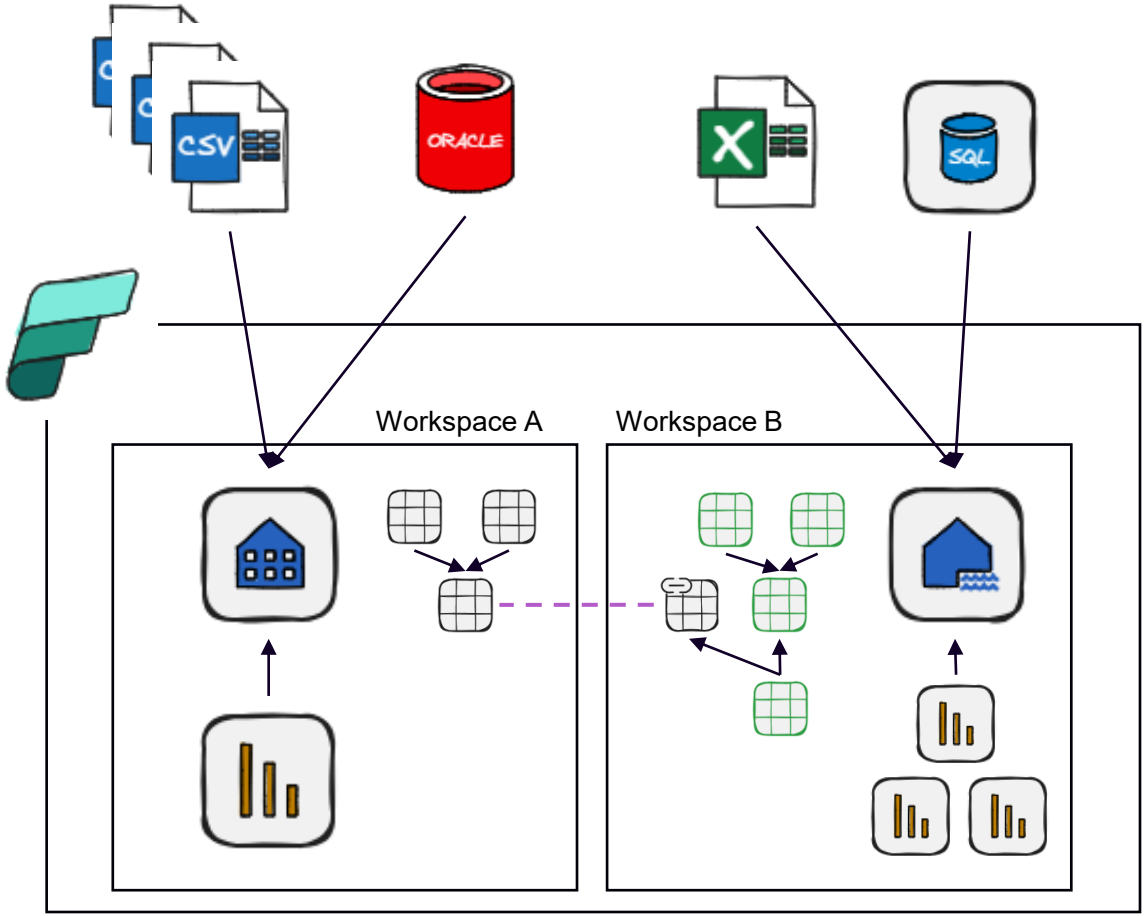
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Doesn’t seem so precious?


NOW (TODAY)
NEW BUILDING, HIDDEN RISK
BEAUTIFUL NEW SKYSCRAPER
BAD GROUND EVEN TODAY!
VERY WEAK SOIL
HIGH WATER TABLE
LOOSE & COMPRESSIBLE SOIL
POSSIBLE VOIDS & SETTLEMENT
LOOKS PERFECT TODAY... BUT THE GROUND IS UNSTABLE!
THEN (FUTURE)
TIME PASSES, PROBLEMS APPEAR
BUILDING STARTS TO TILT
CRACKS DEVELOP
DOORS & WINDOWS DON'T CLOSE PROPERLY
UNEQUAL SETTLEMENT DAMAGES THE STRUCTURE
OVER TIME, THE BAD GROUND CAUSES SERIOUS PROBLEMS AND HIGH COSTS!

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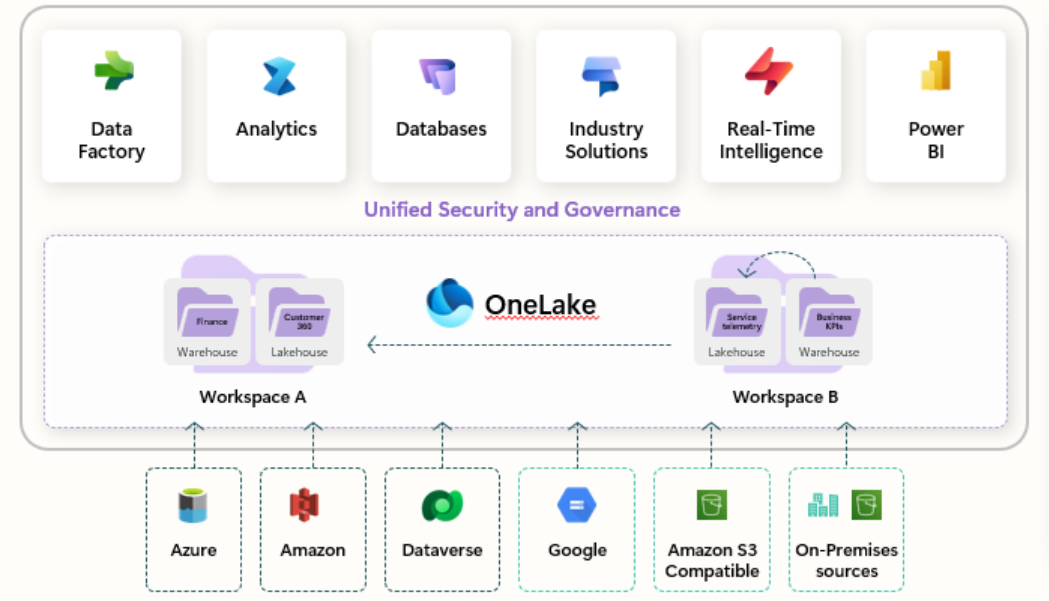
OneLake Shortcuts



Benefits:

1. Space saving
2. Time saving
3. CUs saving
4. Don’t need to reinvent the wheel
5. Data ownership delegated
6. Fix once and for all
7. Add feature once and for all

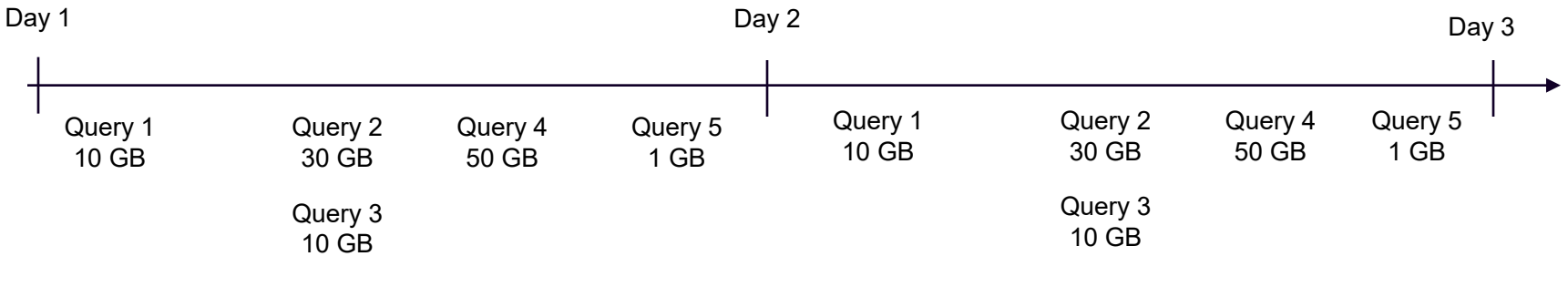
Let’s make the concept bigger



Networking is a factor

- Fabric to Shortcut source path could be long (Italy North -> West US) and **performance** may be impacted
- **Cloud providers charge you** for internet egress

Example on 1 Shortcut reading from AWS*

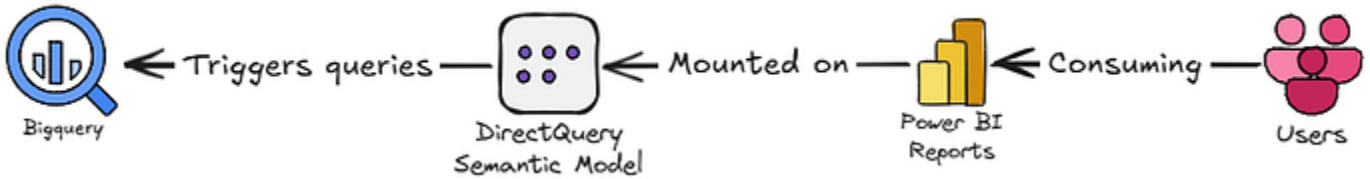


Data Read	10 GB	50 GB	100 GB	101 GB	111 GB	151 GB	201 GB	202 GB
Query Cost	0.00 USD	0.00 USD	0.00 USD	0.09 USD	0.90 USD	3.60 USD	4.50 USD	0.09 USD
Total Bill	0.00 USD	0.00 USD	0.00 USD	0.09 USD	0.99 USD	4.59 USD	9.09 USD	9.18 USD

Day 2 costed 100X Day 1

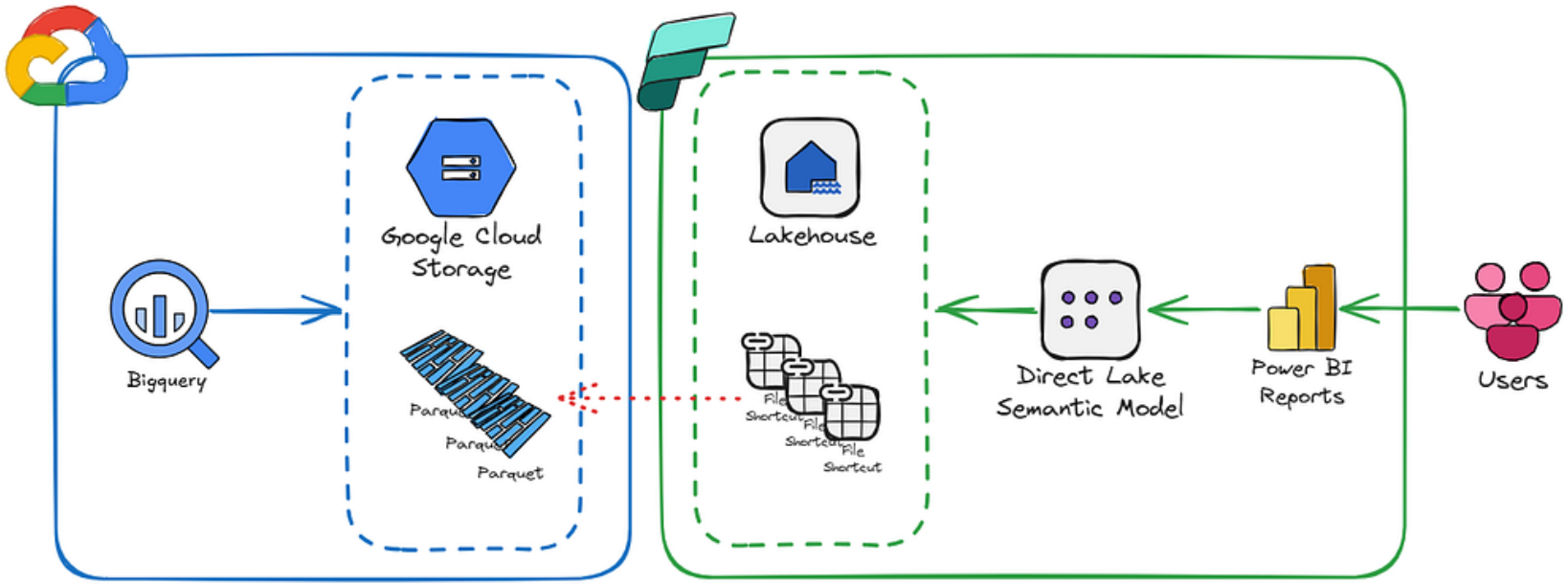
*prices may change

Scenario from the real world



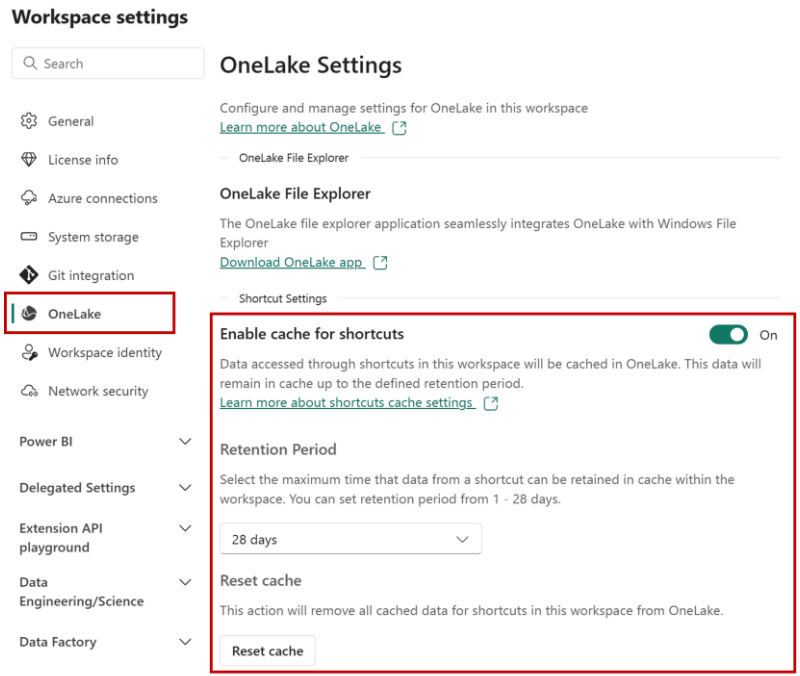
Big query DWH is loaded once per week

We proposed a rebuild



Problem: We're hitting GCS for any query of the user

OneLake caching to reduce Egress



Behind the scenes

- You set the retention between 1-28 days
- Supported:
  - GCS
  - S3
  - S3 Compatible (even on-prem)
- Up to 1GB/file
- First read from source subsequent from Fabric cache unless
  - Retention reached
  - Source has more recent file

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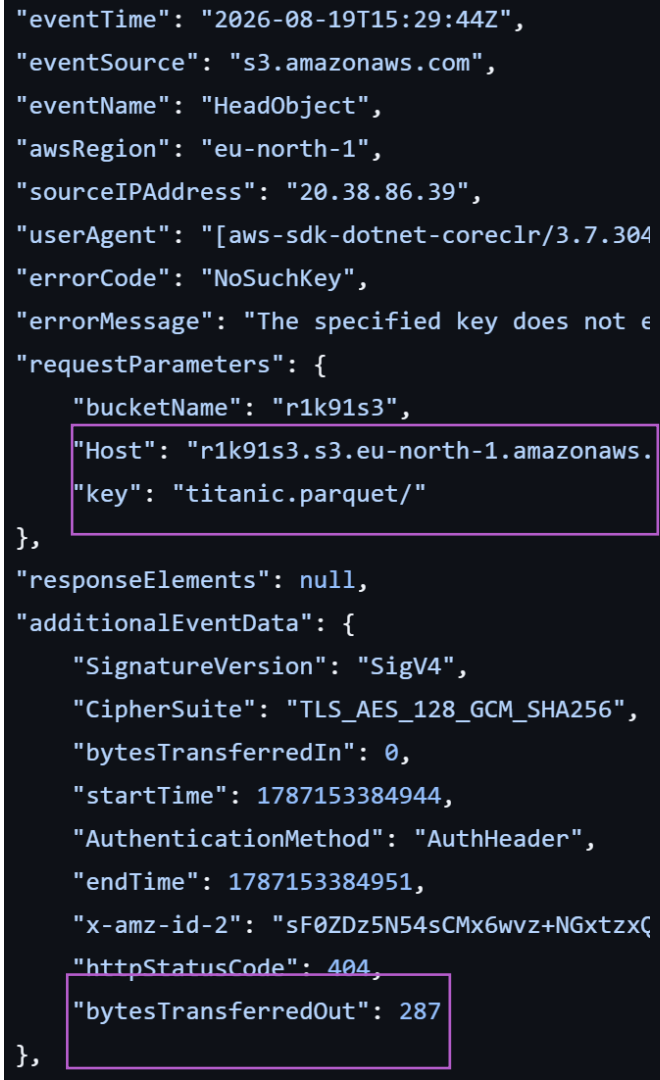
Demo

No Cache

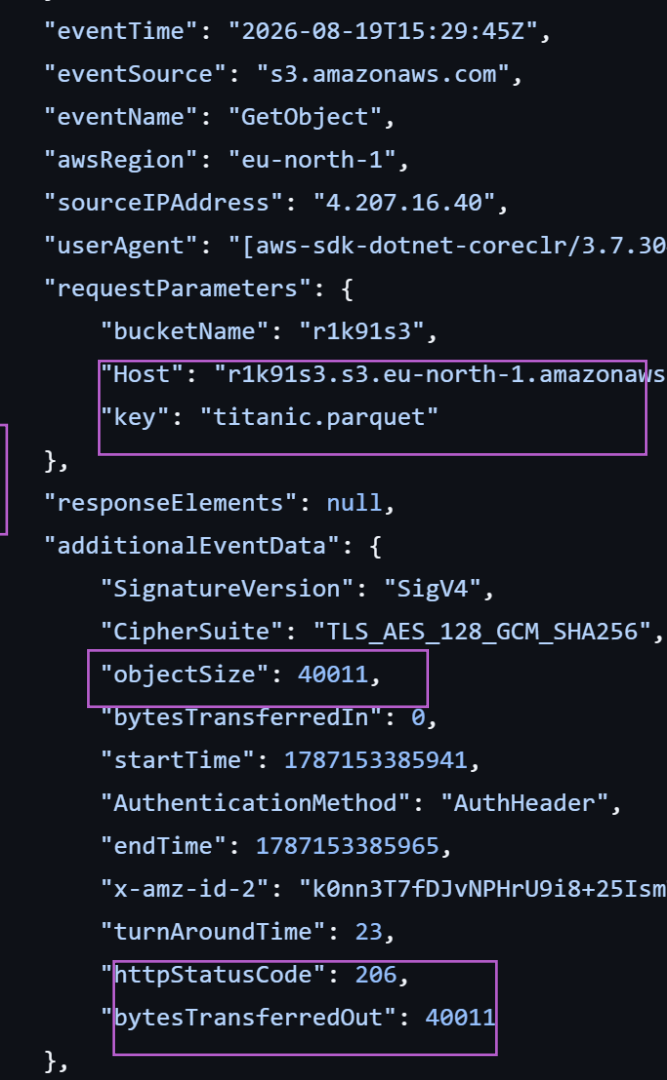
SELECT *

FROM

myLH.dbo.mySC



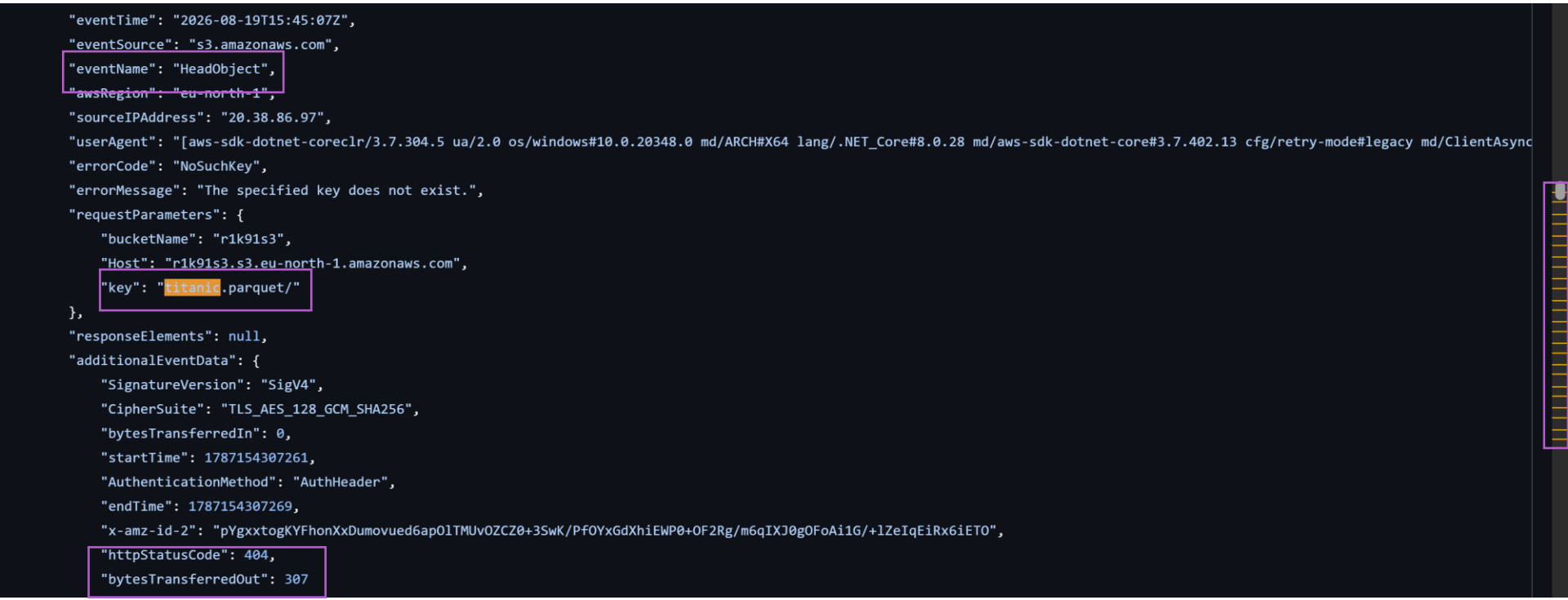
```
"eventTime": "2026-08-19T15:29:44Z",
"eventSource": "s3.amazonaws.com",
"eventName": "HeadObject",
"awsRegion": "eu-north-1",
"sourceIPAddress": "20.38.86.39",
"userAgent": "[aws-sdk-dotnet-coreclr/3.7.304
"errorCode": "NoSuchKey",
"errorMessage": "The specified key does not e
"requestParameters": {
    "bucketName": "r1k91s3",
    "Host": "r1k91s3.s3.eu-north-1.amazonaws.
    "key": "titanic.parquet/"
},
"responseElements": null,
"additionalEventData": {
    "SignatureVersion": "SigV4",
    "CipherSuite": "TLS_AES_128_GCM_SHA256",
    "bytesTransferredIn": 0,
    "startTime": 1787153384944,
    "AuthenticationMethod": "AuthHeader",
    "endTime": 1787153384951,
    "x-amz-id-2": "sF0ZDz5N54sCMx6wvz+NGxtzxC
    "httpStatusCode": 404,
    "bytesTransferredOut": 287
},
```



```
"eventTime": "2026-08-19T15:29:45Z",
"eventSource": "s3.amazonaws.com",
"eventName": "GetObject",
"awsRegion": "eu-north-1",
"sourceIPAddress": "4.207.16.40",
"userAgent": "[aws-sdk-dotnet-coreclr/3.7.30
"requestParameters": {
    "bucketName": "r1k91s3",
    "Host": "r1k91s3.s3.eu-north-1.amazonaws
    "key": "titanic.parquet"
},
"responseElements": null,
"additionalEventData": {
    "SignatureVersion": "SigV4",
    "CipherSuite": "TLS_AES_128_GCM_SHA256",
    "objectSize": 40011,
    "bytesTransferredIn": 0,
    "startTime": 1787153385941,
    "AuthenticationMethod": "AuthHeader",
    "endTime": 1787153385965,
    "x-amz-id-2": "k0nn3T7fDJvNPHrU9i8+25Ism
    "turnAroundTime": 23,
    "httpStatusCode": 206,
    "bytesTransferredOut": 40011
},
```

Cache hit

```
SELECT *
FROM
myLH.dbo.mySC
```

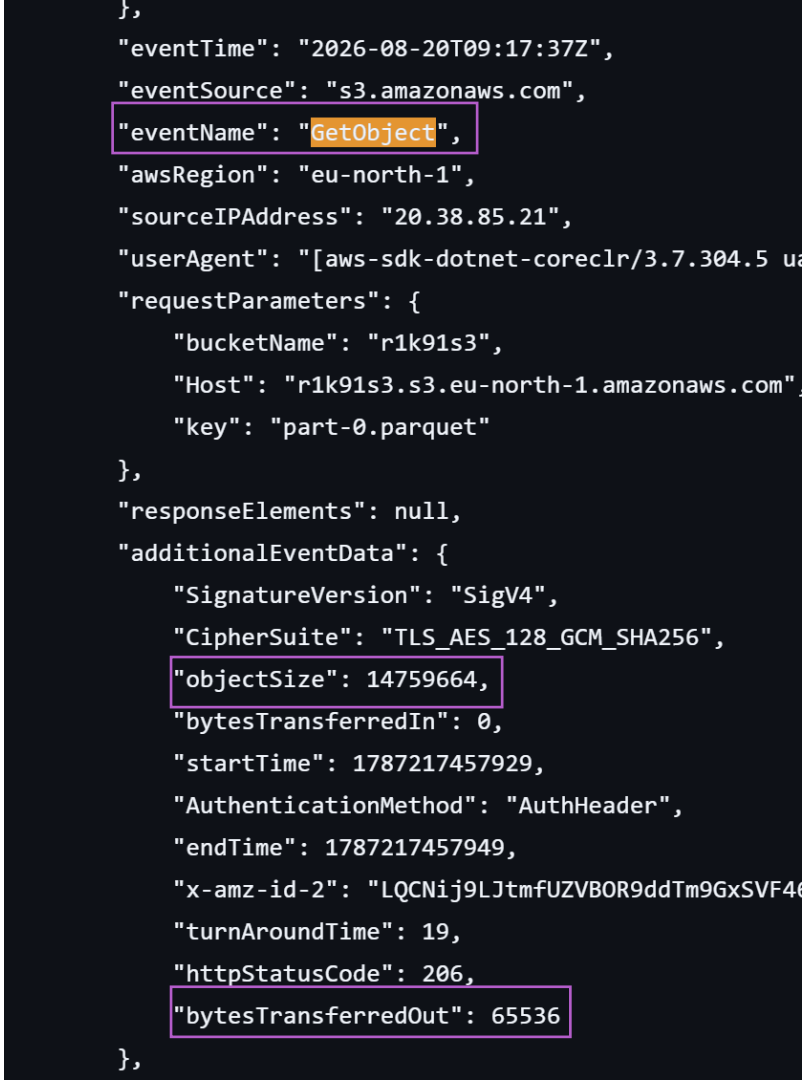


No Cache bigger files

SELECT *

FROM

myLH.dbo.mySC

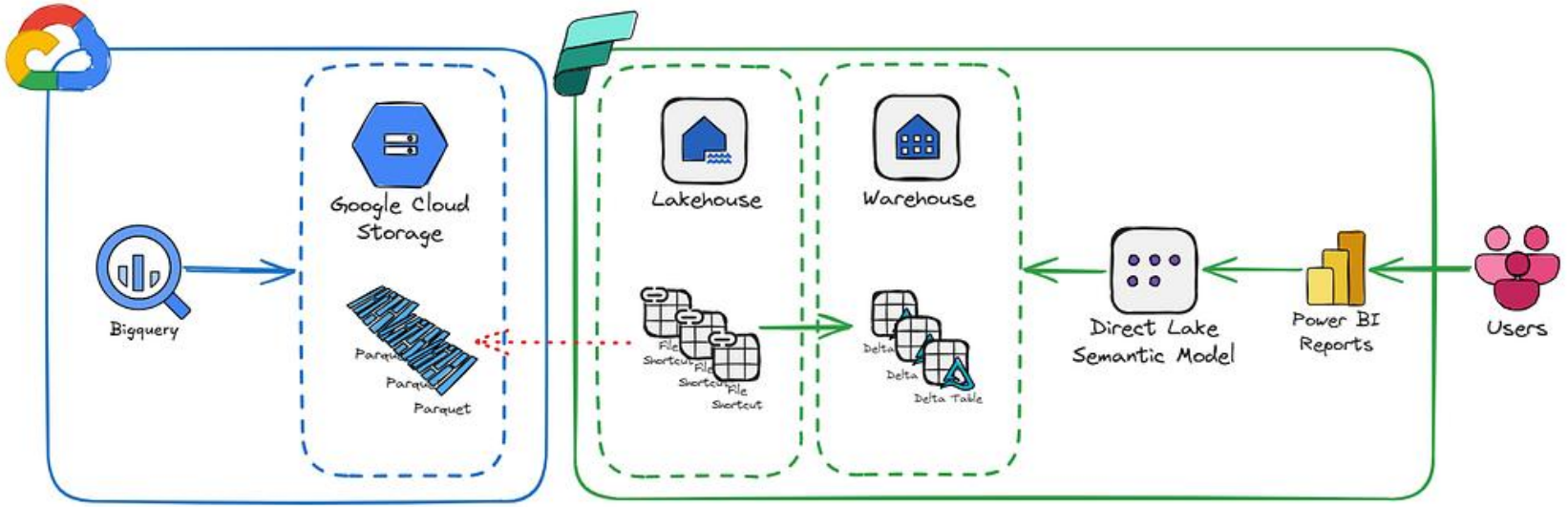


Cache hit on bigger files

```
SELECT *
FROM
myLH.dbo.mySC
```



The final layout



Savings on GC: 92%

Caveats

- Cache observability → you don’t know if a query hit the cache
- Cache visibility → you don’t know if the cache is ready
- Cache scope → at the workspace level

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Q&A

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Thanks!